Advanced E-Commerce Business Intelligence & Risk Analytics System

Project Overview

This project is a complete end-to-end advanced Business Intelligence and Decision Analytics system developed using Python, Pandas, Plotly, Power BI, Power Query, and DAX.

The project was designed to simulate how enterprise-level e-commerce companies use analytics to solve business problems related to profitability, customer retention, operational inefficiencies, return risk, pricing strategy, and marketing performance.

Unlike basic dashboard projects, this solution focuses heavily on:

- business impact
- decision intelligence
- customer behavior analytics
- profitability optimization
- operational risk detection
- executive storytelling

The complete pipeline covered:

- data ingestion
- preprocessing
- feature engineering
- exploratory data analysis
- customer segmentation
- KPI modeling
- dashboard architecture
- business recommendation systems

Business Problem Statement

Modern e-commerce businesses generate massive transactional data but struggle to convert it into actionable decisions.

The company faced multiple critical business challenges:

Business Problem	Business Impact
High discount dependency	Profit margin erosion
Increasing return rates	Revenue leakage and operational loss
Lack of customer segmentation	Inefficient marketing campaigns
No visibility into customer value	Poor retention strategy
High shipping costs	Reduced operational efficiency
Weak traffic sources	Low-quality customer acquisition
No centralized analytics system	Slow decision-making

The primary objective of this project was to build a centralized Business Intelligence ecosystem capable of answering:

- Which customers generate the highest lifetime value?
- Which categories are profitable?
- Which discounts destroy margins?
- Which traffic sources generate risky customers?
- How do shipping costs affect profitability?
- Which products have high return probabilities?
- Where is the business losing money?

Dataset Overview

The dataset consisted of:

- 60,000 transaction records
- 17 original columns
- multi-country customer transactions
- multiple product categories
- customer behavioral attributes
- financial and operational metrics

Raw Dataset Features

Feature	Description
order_id	Unique order identifier
order_date	Transaction timestamp
customer_id	Unique customer identifier
product_id	Product identifier
product_category	Product category
product_price	Original product price
discount_percent	Applied discount percentage
quantity	Number of units purchased
customer_country	Customer geography
traffic_source	Marketing acquisition channel
payment_method	Customer payment mode
shipping_cost	Delivery and logistics cost
rating	Product/customer rating
is_returned	Binary return indicator
discounted_price	Final discounted price
revenue	Revenue generated
profit	Profit earned

Data Cleaning & Preprocessing Pipeline

A complete ETL and preprocessing pipeline was implemented.

Step 1 — Data Loading

- Imported CSV datasets into Python using Pandas

- Validated schema integrity
- Verified record count consistency

Dataset shape:

60,000 rows × 17 columns

Step 2 — Data Type Optimization

Converted:

- order_date → datetime
- monetary columns → decimal
- categorical columns → string/object
- binary columns → integer

This ensured:

- optimized memory usage
- correct aggregations
- accurate time-series analysis

Step 3 — Date Intelligence Engineering

Extracted advanced time-based features:

Feature	Business Use
year	yearly trend analysis
month	seasonality analysis
weekday	weekday performance analysis
day	daily operational analysis
is_weekend	weekend shopping behavior

This enabled:

- trend forecasting
- seasonal analysis
- weekend vs weekday behavior tracking

Step 4 — Data Validation

Validated:

- null values
- duplicate records
- invalid numerical values
- percentage consistency

Ensured data reliability before dashboarding.

Advanced Feature Engineering

One of the strongest components of the project was advanced business-focused feature engineering.

Several analytical columns were engineered to increase insight depth.

Engineered Features

1. Discount Value

discount_value = product_price - discounted_price

Business Impact

Helped quantify:

- exact revenue sacrificed through discounts
- category-level discount dependency

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2. Order Value

order_value = quantity × discounted_price

Business Impact

Enabled:

- order-level revenue analysis
- basket value tracking
- customer spending analysis

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3. Profit Margin

$$\text{Profit Margin} = \frac{\text{Profit}}{\text{Revenue}}$$

Business Impact

Used to:

- identify low-margin products
- detect operational leakage
- evaluate pricing strategy effectiveness

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4. Customer Lifetime Value (CLV)

Customer-level total monetary contribution.

Business Impact

Enabled:

- customer prioritization
- retention strategy
- premium customer identification

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5. Order Frequency

Number of repeat purchases by each customer.

Business Impact

Helped identify:

- loyal customers
- inactive users
- repeat purchase behavior

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6. Recency

Days since last purchase.

Business Impact

Critical for:

- churn detection
- retention campaigns
- reactivation targeting

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7. Weekend Indicator

is_weekend = 1 if Saturday/Sunday else 0

Business Impact

Used to:

- analyze shopping behavior
- identify peak traffic periods
- optimize campaigns

Advanced RFM Customer Segmentation

A complete RFM (Recency, Frequency, Monetary) segmentation framework was implemented.

RFM Model

RFM = (\text{Recency},\ \text{Frequency},\ \text{Monetary})

Customer Segments Created

Segment	Characteristics	Business Meaning
VIP	High frequency + high spending	Highest-value customers
Loyal	Consistent purchases	Stable repeat buyers
Potential	Moderate engagement	Growth opportunity
At Risk	Low recency and low activity	Churn probability

Business Impact of Segmentation

The segmentation model enabled:

- targeted marketing
- customer retention strategies
- personalized campaigns
- churn reduction strategies
- VIP customer prioritization

This transformed the dashboard from descriptive analytics into actionable intelligence.

Exploratory Data Analysis (EDA)

Advanced EDA was performed using Plotly interactive visualizations.

Univariate Analysis

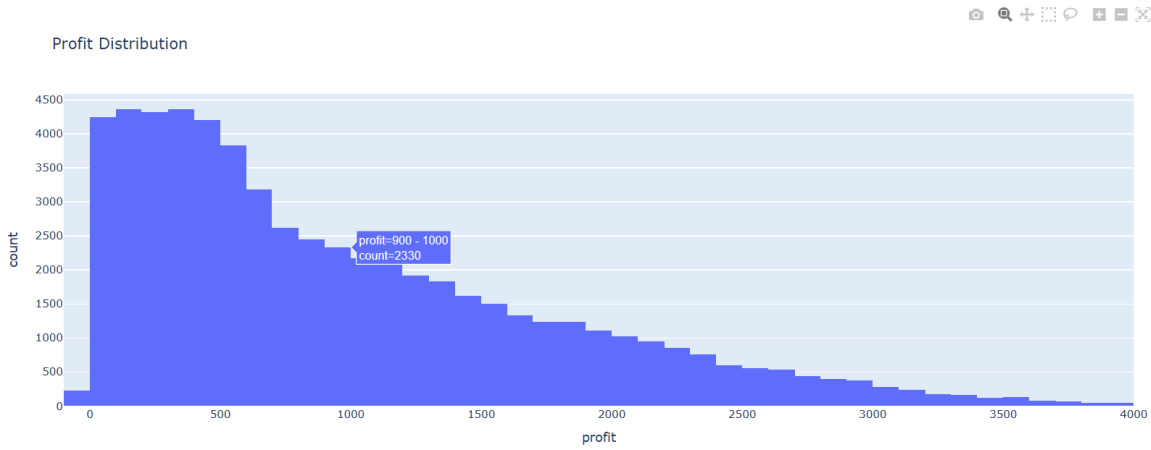
Analyzed:

- revenue distributions
- shipping cost distributions
- product rating distributions
- discount frequency
- category concentration
- traffic source distribution

Key Findings

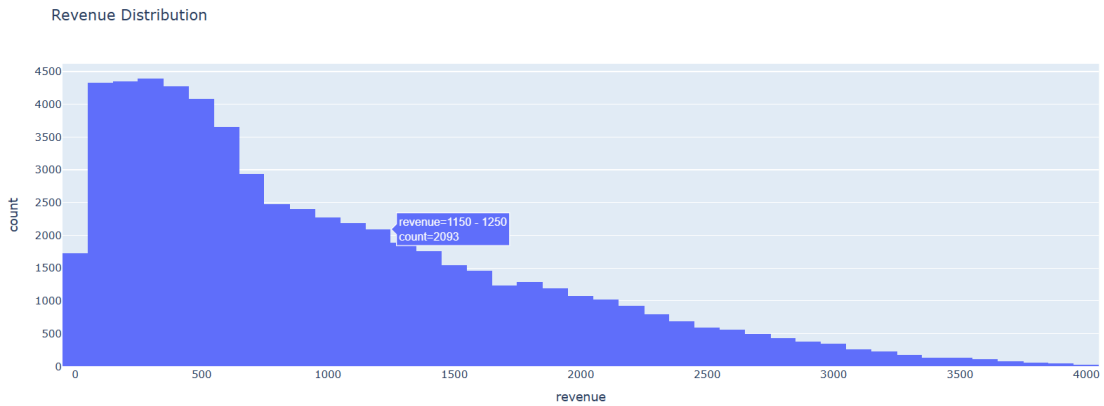
- Revenue distribution showed strong skewness with a few high-value orders driving significant revenue.
- Certain discount levels dominated the dataset.
- Customer ratings clustered around mid-to-high ranges.

```
fig = px.histogram(df, x='profit', nbins=50, title='Profit Distribution')
fig.show()
```

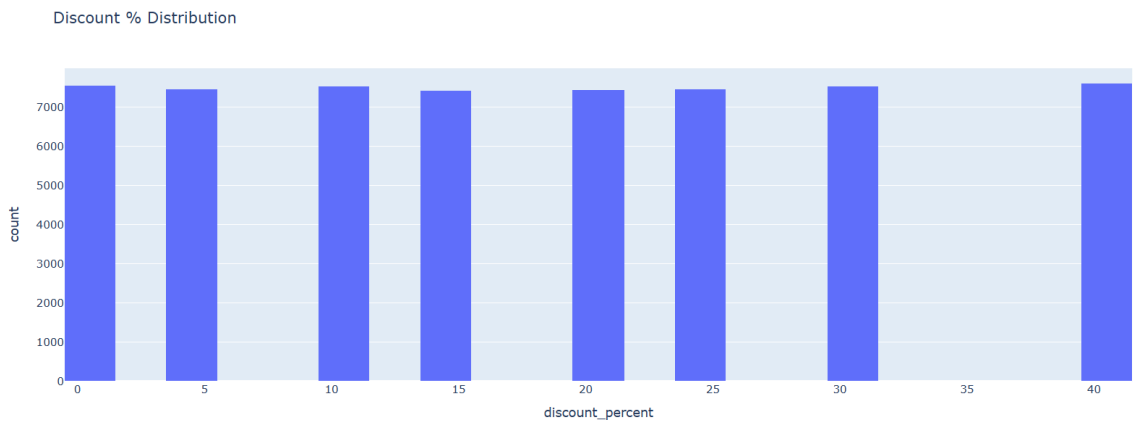


UNIVARIATE ANALYSIS

```
fig = px.histogram(df, x='revenue', nbins=50, title="Revenue Distribution")
fig.show()
```

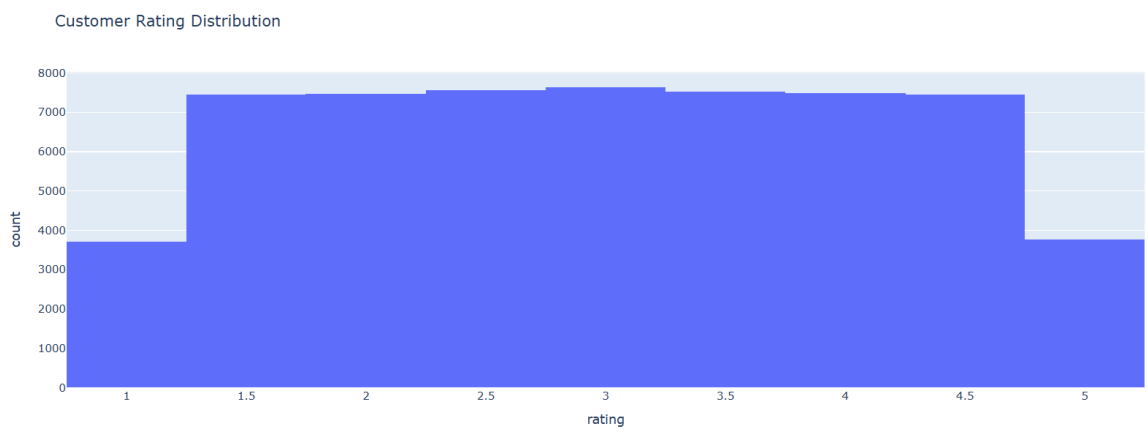


```
fig = px.histogram(df, x='discount_percent', nbins=30, title='Discount % Distribution')
fig.show()
```

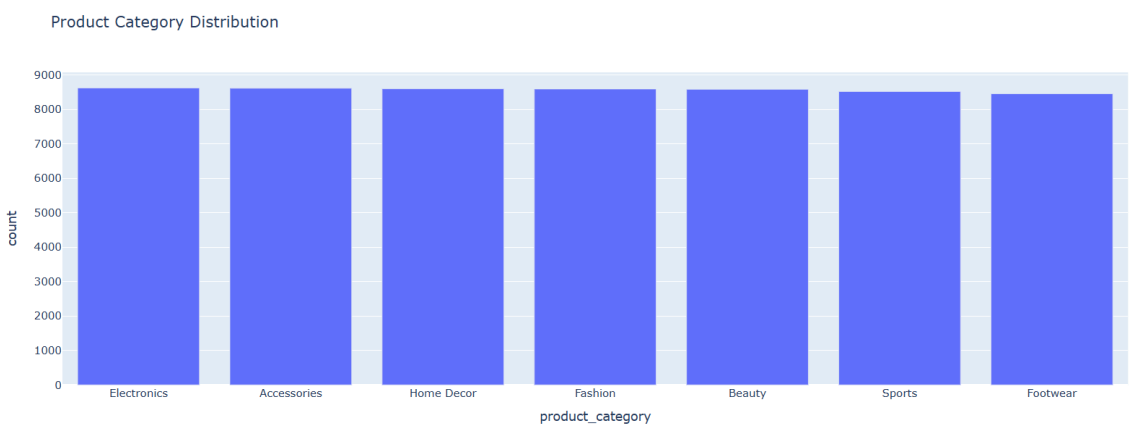


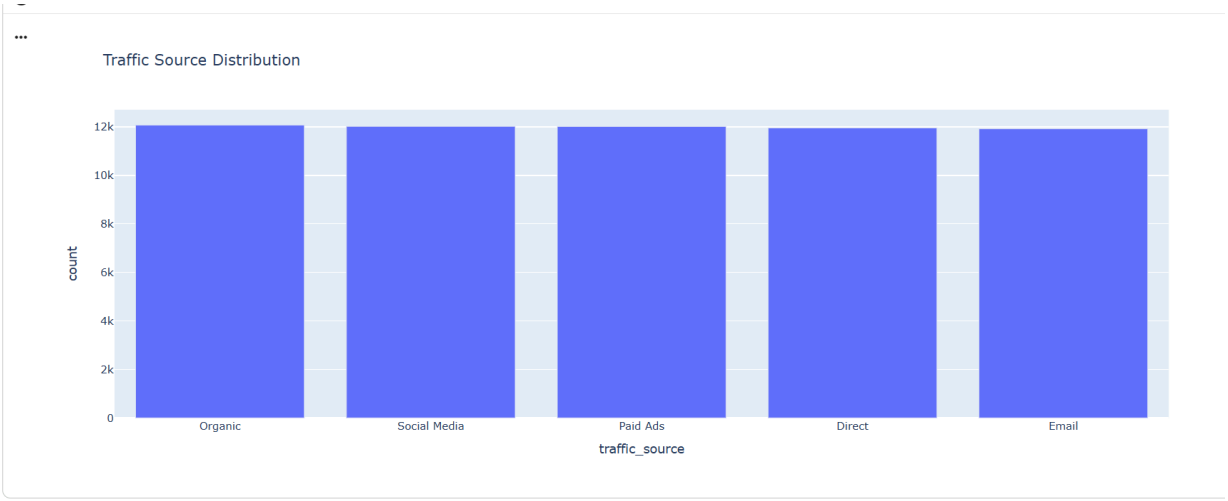
Terminal

```
fig = px.histogram(df, x='rating', nbins=20, title='Customer Rating Distribution')
fig.show()
```




```
fig.show()
```



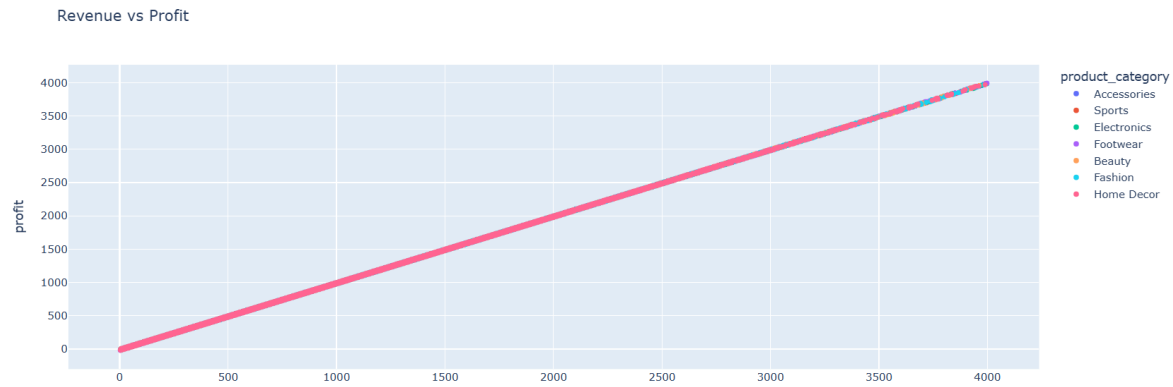


Bivariate & Multivariate Analysis

Analyzed relationships between:

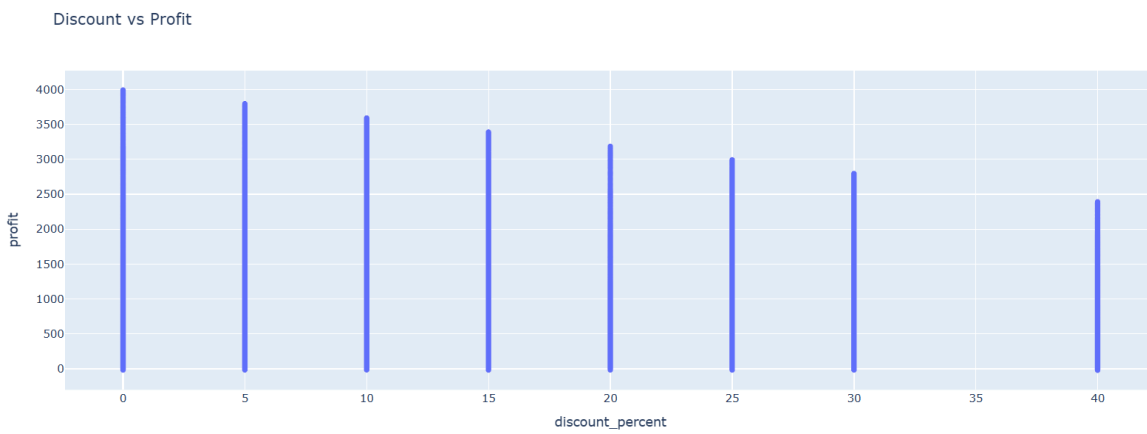
- discount vs profit
- shipping cost vs profitability
- ratings vs return rates
- category vs return behavior
- segment vs CLV
- traffic source vs operational risk

```
fig = px.scatter(df, x='revenue', y='profit',
                 color='product_category',
                 title='Revenue vs Profit')
fig.show()
```

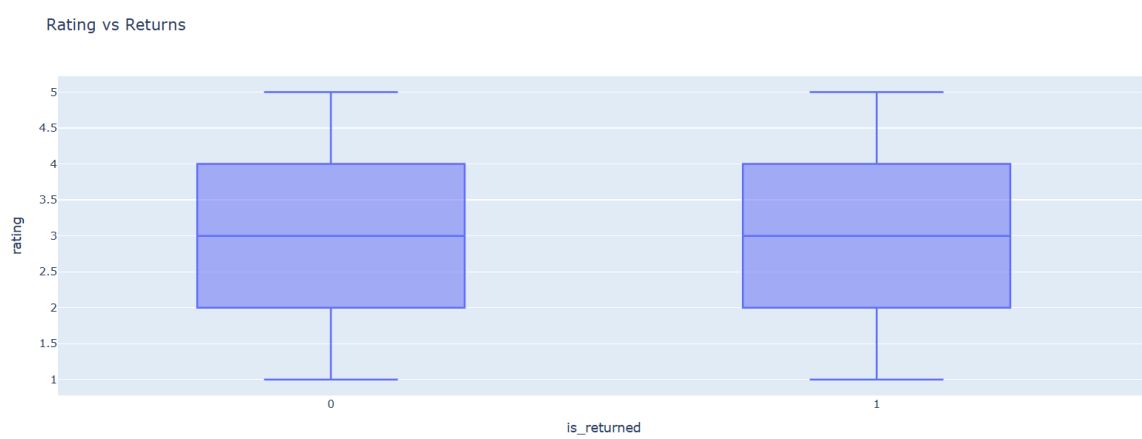


bles Terminal Start

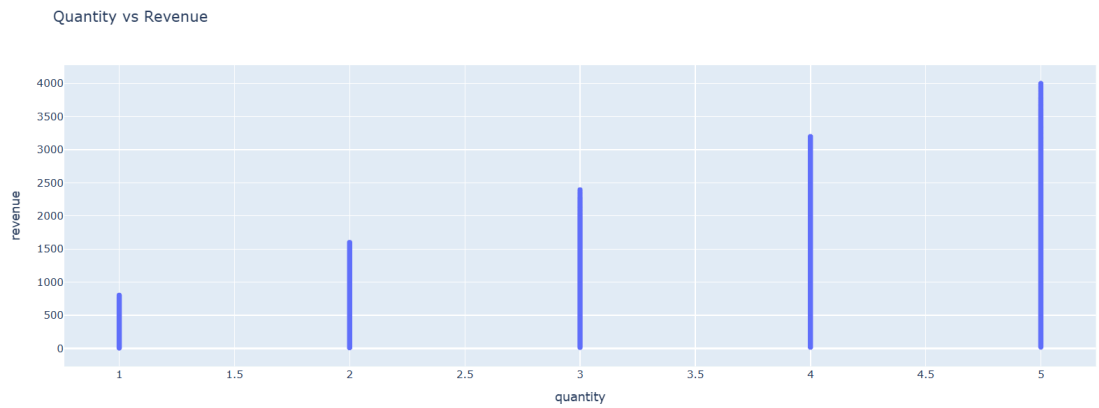
```
fig = px.scatter(df, x='discount_percent', y='profit',
                 title='Discount vs Profit')
fig.show()
```



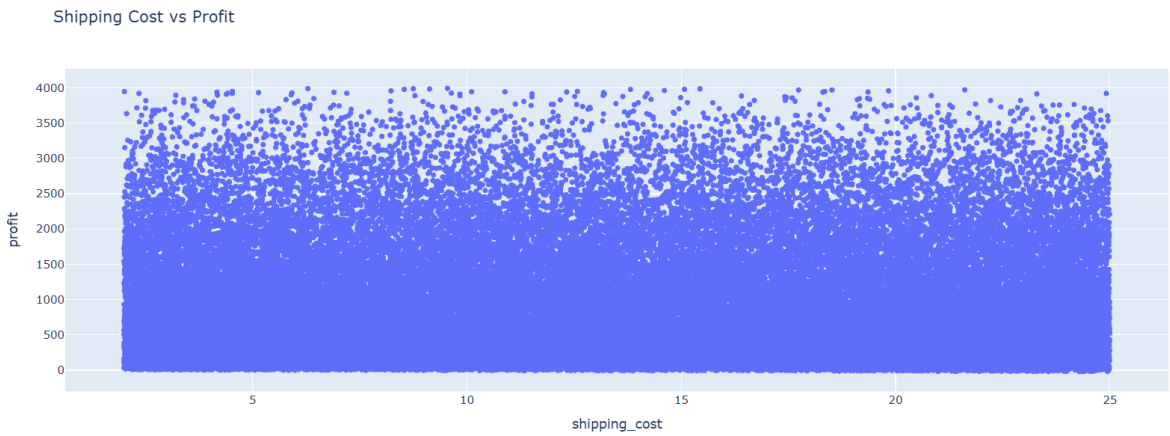

```
fig = px.box(df, x='is_returned', y='rating',
             title='Rating vs Returns')
fig.show()
```



```
fig = px.scatter(df, x='quantity', y='revenue',
                 title='Quantity vs Revenue')
fig.show()
```

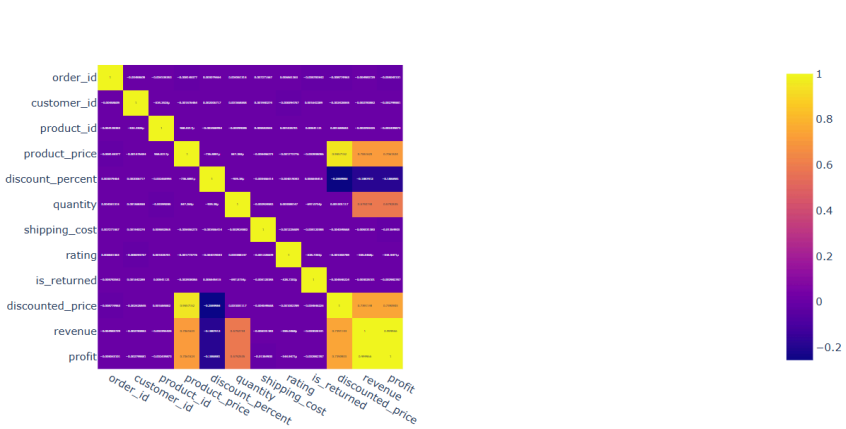



```
fig = px.scatter(df, x='shipping_cost', y='profit',
                 title='Shipping Cost vs Profit')
fig.show()
```

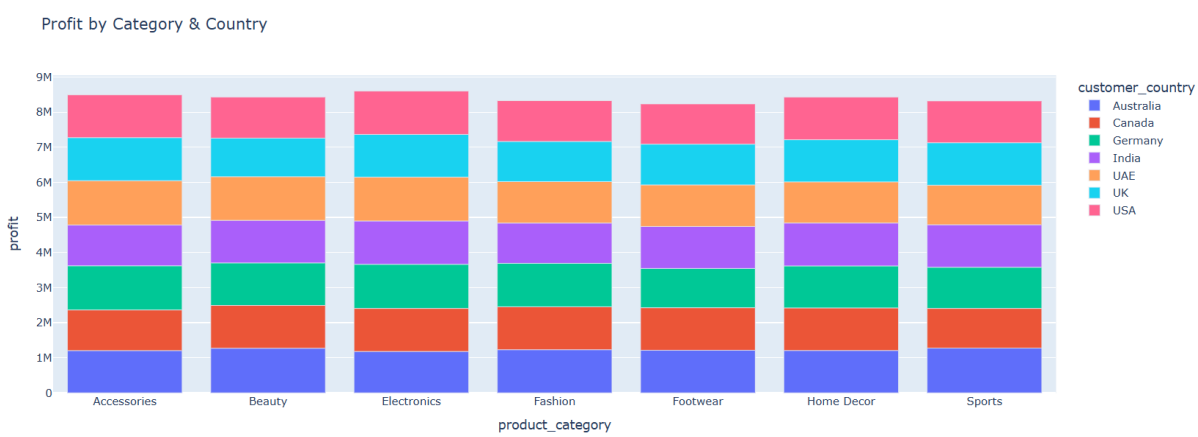


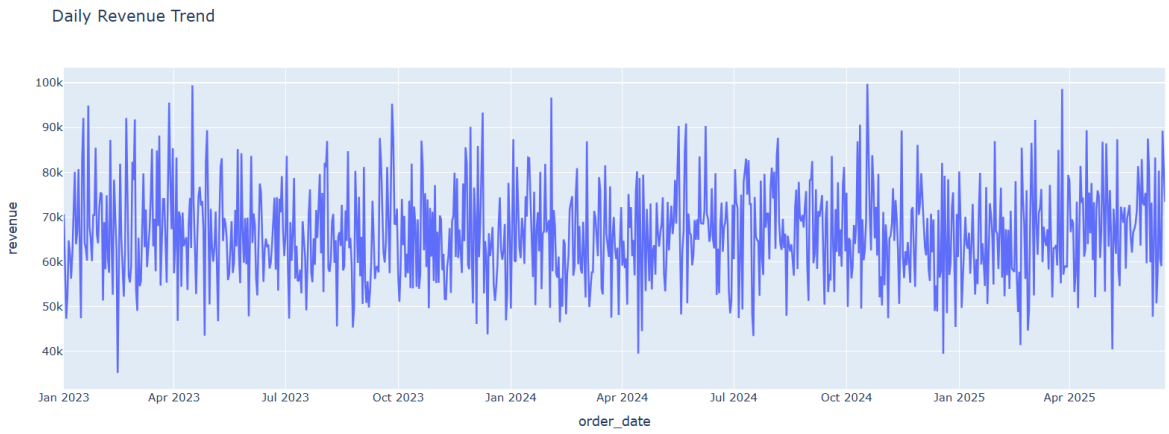
```
fig = px.imshow(corr, text_auto=True, title="Correlation Matrix")
fig.show()
```

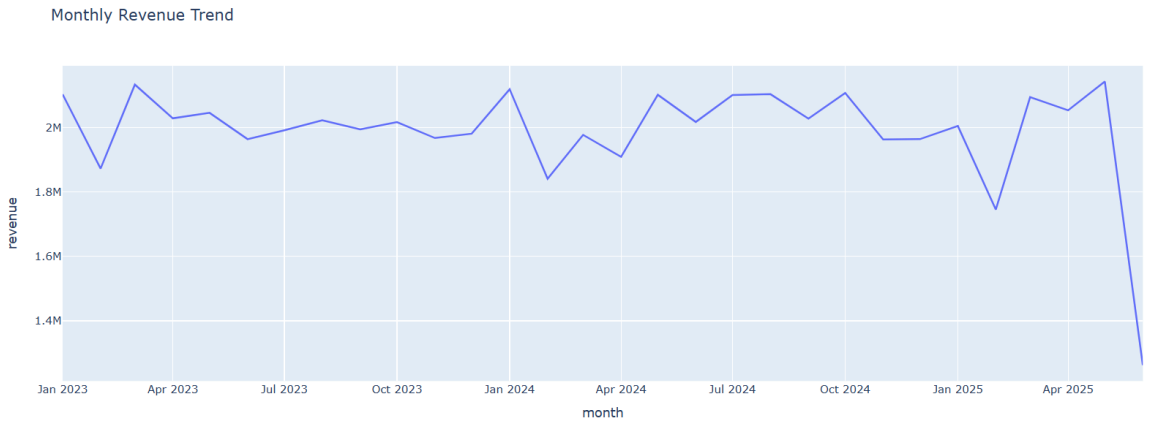
...



Terminal







Major Analytical Findings

Discount vs Profitability

High discounts frequently reduced overall profitability despite increasing order volume.

Business Impact

Showed that:

- aggressive discounting was unsustainable
- revenue growth did not always translate into profit growth

Ratings vs Return Rate

Lower-rated products consistently showed higher return rates.

Business Impact

Suggested:

- product quality issues
- customer dissatisfaction
- inaccurate product expectations

Shipping Cost vs Profit

Higher shipping costs negatively impacted profit margins.

Business Impact

Revealed:

- operational inefficiencies
- logistics optimization opportunities

Traffic Source Quality Analysis

Certain traffic sources generated:

- higher return rates
- lower profitability
- weaker customer retention

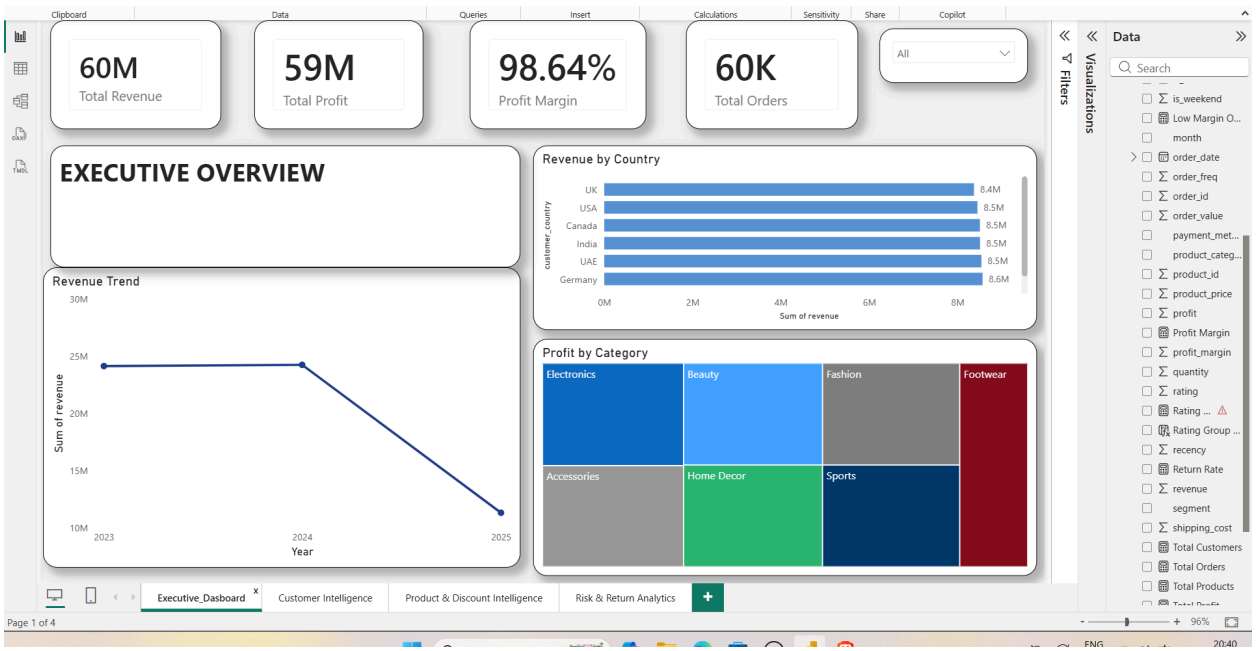
Business Impact

Showed inefficient marketing acquisition channels.

Power BI Dashboard Architecture

The final solution was divided into four major executive dashboards.

1. Executive Overview Dashboard



Objective

Provide high-level business performance overview.

KPIs Implemented

KPI	Business Meaning

Total Revenue	Total business earnings
Total Profit	Net profitability
Profit Margin	Business efficiency
Total Orders	Operational scale

Visualizations Used

- Revenue trend analysis
- Country-level revenue analysis
- Category profitability treemap
- Interactive slicers

Key Insights

- Revenue remained strong overall but showed declining trends during certain periods.
- Some categories generated high revenue but low profit margins.
- Revenue concentration existed in a few major countries.

Business Recommendations

- Focus on margin optimization instead of pure revenue growth.
- Expand in high-performing geographical regions.
- Reduce dependency on low-margin product categories.

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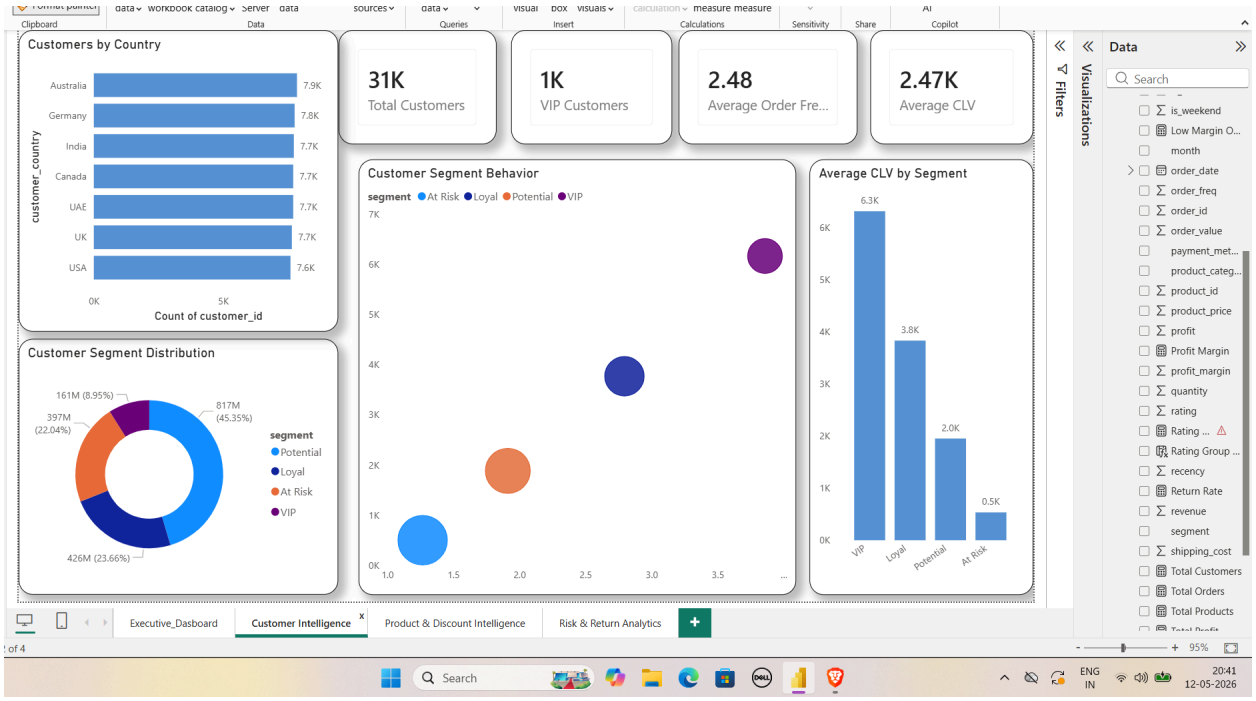
2. Customer Intelligence Dashboard



Objective

Analyze customer behavior and lifetime value.

KPIs Implemented

KPI	Purpose
Total Customers	Customer base size
VIP Customers	Premium segment tracking
Average CLV	Customer value analysis
Average Order Frequency	Retention analysis

Visualizations Used

- RFM segmentation donut chart
- Customer behavior scatter plots
- CLV analysis
- Country-level customer concentration

Key Insights

- VIP customers contributed disproportionately high revenue.
- At-risk customers had significantly lower purchase frequency.
- Loyal customers maintained stable repeat purchase behavior.

Business Recommendations

VIP Customers

- loyalty programs
- premium memberships
- exclusive product access

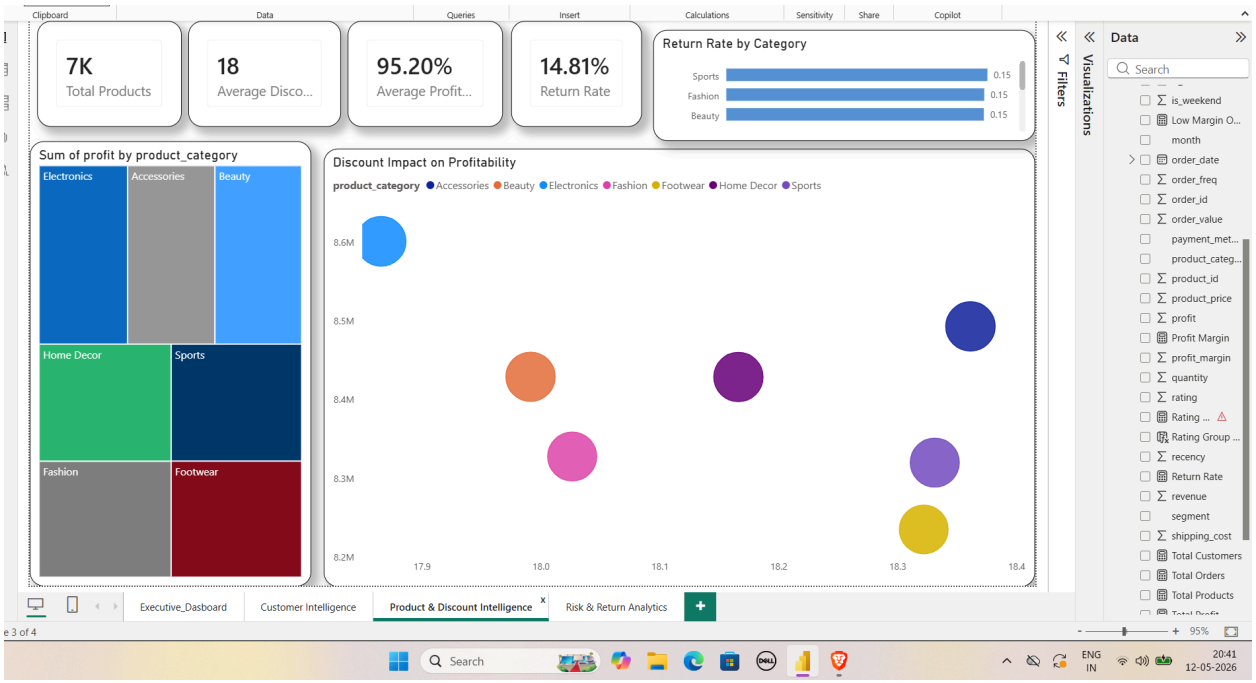
At-Risk Customers

- personalized remarketing
- reactivation campaigns
- retention discounts

Loyal Customers

- cross-selling
- upselling
- referral programs

3. Product & Discount Intelligence Dashboard



Objective

Evaluate pricing strategy and category profitability.

KPIs Implemented

KPI	Purpose
Total Products	Product scale
Average Discount	Pricing intensity

Average Profit Margin	Margin performance
Return Rate	Product risk

Visualizations Used

- Discount vs profitability scatter plots
- Product category treemap
- Return rate analysis
- Shipping impact analysis

Key Insights

- Higher discounts frequently reduced profit margins.
- Certain categories sustained profitability despite moderate discounts.
- Some categories showed extremely high return rates.

Business Recommendations

Pricing Strategy

- optimize discounts category-wise
- avoid blanket discounting

Product Strategy

- prioritize high-margin categories
- reduce exposure to risky products

Return Reduction

Improve:

- product quality
- descriptions
- customer expectation management

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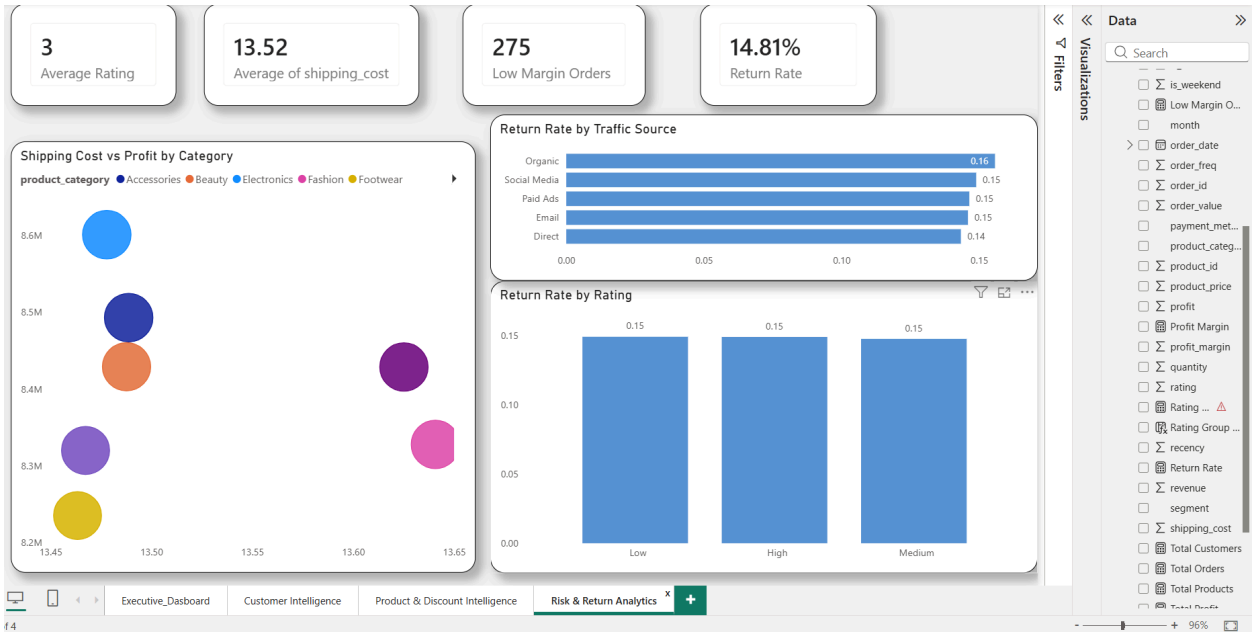
4. Risk & Return Analytics Dashboard



Objective

Identify operational and profitability risks.

KPIs Implemented

KPI	Purpose
Return Rate	Operational risk
Avg Shipping Cost	Logistics efficiency

Avg Rating	Customer satisfaction
Low Margin Orders	Margin leakage

Visualizations Used

- Return rate by traffic source
- Shipping cost vs profitability
- Rating vs return behavior
- Operational risk analysis

Key Insights

- Lower-rated products had significantly higher return probabilities.
- Some traffic sources generated low-quality customers.
- Shipping costs created direct pressure on profitability.

Business Recommendations

Logistics Optimization

- optimize warehouse distribution
- reduce delivery inefficiencies

Marketing Optimization

- reduce investment in poor-performing channels
- focus on high-conversion sources

Product Improvement

- prioritize low-rated categories
- monitor customer feedback continuously

Technical Stack

Technology	Purpose
Python	Data preprocessing
Pandas	Data manipulation
Plotly	Interactive EDA
Power BI	Dashboard development
DAX	KPI calculations
Power Query	ETL workflows

Advanced DAX Measures

Custom DAX measures included:

- Profit Margin
- Return Rate
- Average CLV
- Customer Frequency
- Low Margin Orders
- Revenue Intelligence KPIs

Business Value Delivered

This project demonstrates how advanced analytics can help businesses:

- improve profitability
- reduce operational losses
- optimize pricing strategy
- improve customer retention
- identify churn risk
- improve logistics efficiency
- optimize marketing spend
- improve strategic decision-making

Final Conclusion

This project transformed raw transactional data into a complete enterprise-level Business Intelligence ecosystem.

The final system provides:

- executive visibility
- customer intelligence
- operational risk analysis
- profitability optimization
- strategic decision support

The project reflects real-world analytical workflows used in modern data-driven organizations and demonstrates strong capabilities in:

- analytics engineering
- dashboard storytelling
- business intelligence
- customer analytics
- decision intelligence
- operational analytics
- strategic business recommendations